PROJECT DESCRIPTION REPORT

System Title: Hospital Management System

1. Project Title and Business Idea

Project Title: Hospital Management System (HMS).

Business Idea: A comprehensive web-based or desktop information system designed for healthcare institutions to streamline hospital operations, manage patient records, coordinate doctor schedules, process medical

appointments, and handle patient billing efficiently.

2. Problem Statement and Purpose of the Proposed System

Problem Statement: Traditional or manual hospital record-keeping often leads to scheduling conflicts, misplacement of patient medical histories, delayed billing processes, and difficulties in tracking doctor availabilities across different departments.

Purpose of the Proposed System: The primary purpose of this system is to automate core hospital workflows, reduce human error, provide quick and secure access to patient profiles, streamline appointment bookings, and ensure accurate financial tracking through integrated billing.

3. Project Scope

Functions Covered (In-Scope):

Patient registration, profile updates, and medical history tracking.

Doctor management, specialization tracking, and schedule viewing.

Department management and organization of medical staff.

Appointment booking, confirmation, and cancellation processing.

Bill generation and payment status processing for patients.

Functions Excluded (Out-of-Scope):

Advanced pharmacy inventory management and drug supply chain logistics.

Electronic health record (EHR) interoperability with external national health networks.

Full physical implementation and deployment; the project focuses strictly on software analysis, requirements specification, and UML class modeling.

4. Main Classes and Their Responsibilities:

Based on the system design, the main classes and their core responsibilities include:

Patient: Represents individuals seeking medical care. Responsible for storing personal details

(patientID, fullName, age, gender, phoneNumber) and handling registration, profile updates, and booking

appointments. assignment_UML_class... +1

Doctor: Represents medical professionals working within the hospital. Stores professional data (doctorID, fullName, specialization , email) and manages schedules, patient diagnoses, and prescription writing.

Appointment: Represents the scheduling link between patients and doctors. Manages booking details (appointmentID, appointmentDate, st atus) and handles confirmations or cancellations.

Department: Represents administrative or medical units in the hospital (e.g., Cardiology, Pediatrics). Manages department identification (deptID, deptName) and doctor assignments.

Bill: Represents financial transactions for medical services rendered. Manages payment details (billID, totalAmount, paymentStatus) and processes billing generation and payments.

5. Brief Description of the System Requirements:

• Functional Requirements: The system must allow patients to register and book appointments, permit doctors to view schedules and diagnose patients, enable department heads to assign medical staff, and generate automated bills upon service completion.

Non-Functional Requirements: The system must maintain data privacy and security for sensitive patient medical records, ensure high availability for appointment scheduling, and provide a user-friendly interface for both administrative staff and patients.